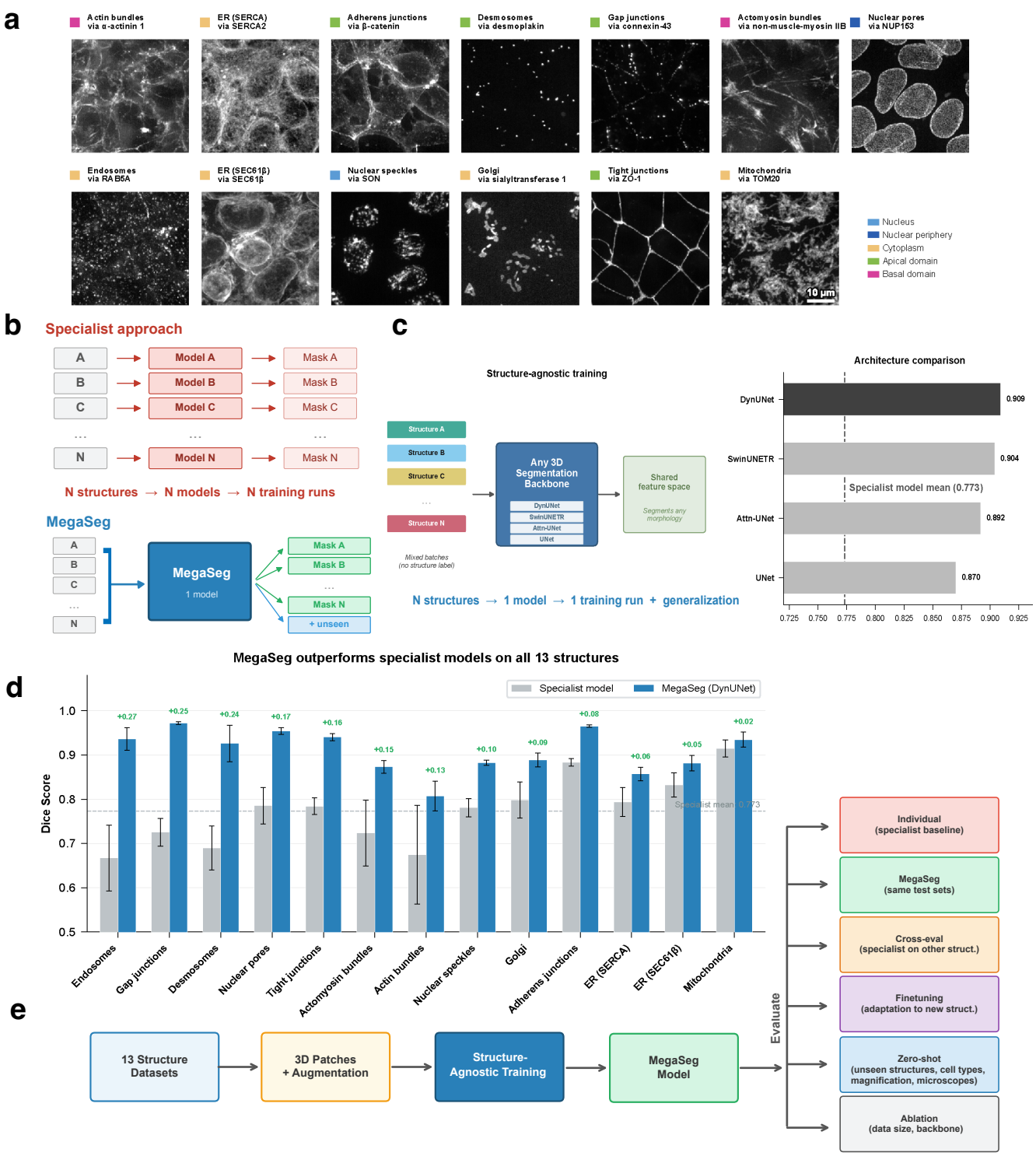

a
Actin bundles via α-actinin 1
ER (SERCA) via SERCA2
Adherens junctions via β-catenin
Desmosomes via desmoplakin
Gap junctions via connexin-43
Actomyosin bundles via non-muscle-myosin IIB
Nuclear pores via NUP153
Endosomes via RAB5A
ER (SEC61β) via SEC61β
Nuclear speckles via SON
Golgi via sialyltransferase 1
Tight junctions via ZO-1
Mitochondria via TOM20
10 µm
Nucleus
Nuclear periphery
Cytoplasm
Apical domain
Basal domain
b
Specialist approach
A
B
C
N
Model A
Model B
Model C
Model N
Mask A
Mask B
Mask C
Mask N
N structures → N models → N training runs
MegaSeg
MegaSeg
1 model
Mask A
Mask B
Mask N
+ unseen
c
Structure-agnostic training
Structure A
Structure B
Structure C
Structure N
Mixed batches (no structure label)
Any 3D Segmentation Backbone
DynUNet
SwinUNETR
Attn-UNet
UNet
Shared feature space
Segments any morphology
N structures → 1 model → 1 training run + generalization
Architecture comparison
DynUNet 0.909
SwinUNETR 0.904
Specialist model mean (0.773)
Attn-UNet 0.892
UNet 0.870
0.725 0.750 0.775 0.800 0.825 0.850 0.875 0.900 0.925
d
MegaSeg outperforms specialist models on all 13 structures
Specialist model
MegaSeg (DynUNet)
Dice Score
1.0 0.9 0.8 0.7 0.6 0.5
+0.27 +0.25 +0.24 +0.17 +0.16 +0.15 +0.13 +0.10 +0.09 +0.08 +0.06 +0.05 +0.02
Specialist mean 0.773
Endosomes
Gap junctions
Desmosomes
Nuclear pores
Tight junctions
Actomyosin bundles
Actin bundles
Nuclear speckles
Golgi
Adherens junctions
ER (SERCA)
ER (SEC61β)
Mitochondria
e
13 Structure Datasets
3D Patches + Augmentation
Structure-Agnostic Training
MegaSeg Model
Evaluate
Individual (specialist baseline)
MegaSeg (same test sets)
Cross-eval (specialist on other struct.)
Finetuning (adaptation to new struct.)
Zero-shot (unseen structures, cell types, magnification, microscopes)
Ablation (data size, backbone)